

Paper_ED_Cos_04 — Linear Structure Formation as Substrate-Inherited Growth of Primordial Perturbations

Series: Wave-3 Generative Papers (Cosmology Arc — linear structure-formation theory)
Status: Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Cos_01 (primordial spectrum) + Cos_02 (baryon density) + Cos_03 (acoustic physics + transfer-function content) + Cos_05 (late-time expansion) + Route A4 substrate-parameter inheritance + standard cosmological-perturbation-theory machinery via DCGT continuum bridge. **M3 via continuum-Friedmann inheritance** — Dyn_02 M2 RDE/MDE chain-class identification is *not* load-bearing (see Preamble item 6). **Date:** 2026-05-17 **Authors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_012 (substrate-c); Paper_027 (Newton's G); Paper_028 ($R_H = c/H_0$); Paper_ED_SC_4_9 (S1/S2/S3); Paper_ED_Cos_01 (M3 — Inflation); Paper_ED_Cos_02 (M3 — BBN); Paper_ED_Cos_03 (M3 — CMB acoustic); Paper_ED_Cos_05 (M3 — Dark Energy); Paper_ED_Dyn_02 (M2 — RDE/MDE, upstream-status only); Paper_095 (verdict grammar — §3.2.1 sub-labels). Substrate-research support memos cross-referenced in §2.

Label-class note: Audit-row sub-labels *substrate-parameter-INHERITED* and *IC-INHERITED* per Paper_095 §3.2.1.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim closed-proof derivation of standard cosmological-perturbation-theory machinery (Newtonian/relativistic perturbation theory; CAMB / CLASS Boltzmann codes for transfer functions; halofit / EFTofLSS for nonlinear extensions) from substrate primitives. These continuum-side machineries are **INHERITED** via DCGT continuum bridge; ED supplies the substrate-graph reading of $H(t)$ + primordial-spectrum inheritance underlying them.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program. Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The structure-formation content is composition of Cos_01 + Cos_02 + Cos_03 + Cos_05 + Route A4 + standard cosmological-perturbation-theory + standard Boltzmann codes via DCGT continuum bridge.
4. It does **not** claim numerical-precision derivation of specific observables ($\sigma_8 \approx 0.81$; matter density $\Omega_m h^2 \approx 0.143$; BAO scale $r_d \approx 147$ Mpc; galaxy-survey $P(k)$). These inherit standard CAMB / CLASS Boltzmann-code outputs + standard galaxy-survey analysis pipelines.
5. It does **not** address nonlinear structure formation (halofit; EFTofLSS; full N-body simulations) at substrate-graph derivational level. Nonlinear regime is standard astrophysics + cosmological-simulation territory; ED supplies the linear-theory substrate-graph framework only.
6. **Inheritance from Dyn_02 M2 status (architectural division).** Cos_04 inherits standard-cosmology continuum-Friedmann $H(t)$ + standard linearized cosmological-perturbation theory via DCGT continuum bridge at M3-equivalent inheritance level. Dyn_02's substrate-graph RDE / MDE chain-class identifications (M2 with OPEN-HM-Q1 + OPEN-HM-Q2) remain OPEN at Dyn_02's level; **Cos_04 does not load-bear on**

these closures because the linear-structure substrate-graph content (Route A4 $H_0 + \Theta_{\text{ED}}$ sensitivity channel for $D(a)$, $T(k)$, r_s , σ_8) is independent of the chain-class identification for radiation/matter content. Same architectural-division logic as Cos_02 + Cos_03 + Dyn_01 + Dyn_04 + GW_01.

Abstract

Linear structure formation is identified at substrate-graph level as **the substrate-inherited solution to the linearized cosmological-perturbation-theory equations on the ED expansion-history background**. Growth factor $D(a)$ inherits substrate-graph $H(t)$ via continuum-Friedmann inheritance (Cos_01 + Cos_05 + Route A4); transfer function $T(k)$ inherits Cos_03 acoustic physics + Cos_02 baryon density via k_{eq} scale; primordial spectrum $P_{\text{init}}(k) \propto k^{n_s-1}$ with $n_s \approx 0.965$ inherits Cos_01; full matter power spectrum $P(k, a) = D^2(a)T^2(k)P_{\text{init}}(k)$ inherits standard cosmological-perturbation theory via DCGT continuum bridge; H_0 at substrate-parameter-INHERITED level via Route A4. Three growth regimes: radiation-era Meszaros suppression at $z > z_{\text{eq}} \approx 3400 \rightarrow$ matter-era growth $D \propto a \rightarrow$ late-LDE growth suppression at $z < z_\Lambda \approx 0.7$. BAO feature at $r_s \approx 147$ Mpc inherits Cos_03. Substrate-c preserved. **Form-IDENTIFIED** via Cos_01 + Cos_02 + Cos_03 + Cos_05 + DCGT. **Value-INHERITED**: $D(a)$, $T(k)$, r_s , σ_8 via CAMB / CLASS + Route A4. **IC-INHERITED**: primordial spectrum (Cos_01), matter/baryon densities (Cos_02 + Cos_03), recombination physics. **Verdict per Paper_095: M3** (form-IDENTIFIED + value-INHERITED + IC-INHERITED); Dyn_02 M2 *not* load-bearing (Preamble item 6). **F1 (sharpest)**: matter power-spectrum $P(k)$, BAO scale, or $f\sigma_8(z)$ significantly inconsistent with substrate-graph composition under Route A4 scaling — beyond galaxy-survey + CMB-lensing precision — refutes the composition. **Empirical anchors are exceptionally strong**: BOSS / eBOSS / DESI BAO (sub-percent precision); DES / KiDS / HSC weak-lensing; Planck CMB-lensing; future Euclid / LSST / Roman. **Distinguishing ED contribution**: Route A4 substrate-parameter Θ_{ED} scaling channel for $D(a)$, $T(k)$, r_s , σ_8 supplies ED's structural link to the $S_8 \equiv \sigma_8\sqrt{\Omega_m}/0.3$ tension observational target (~ 3 between CMB and weak-lensing); whether the tension resolves via substrate-parameter tightening (RA-OPEN-4a) or standard intermediate-redshift physics modification is OPEN and not load-bearing for Cos_04's M3 verdict. Completes Cosmology Arc linear-theory coverage through dark-energy era; enables Paper_ED_Cos_06 (Inflationary fluctuation spectrum).

§1 Statement of Result

Linear structure formation is the substrate-graph **growth of primordial density perturbations** $\delta(k, a) = D(a) \cdot T(k) \cdot \delta_{\text{init}}(k)$ on the ED cosmological-history background, with matter power spectrum:

$$P(k, a) = D^2(a) \cdot T^2(k) \cdot P_{\text{init}}(k), \quad P_{\text{init}}(k) \propto k^{n_s-1}, \quad n_s \approx 0.965$$

where $D(a)$ is the growth factor (substrate-graph $H(t)$ inheritance from Cos_05 + Cos_01), $T(k)$ is the transfer function (Cos_03 acoustic-physics + matter-radiation-equality scale inheritance), and $\delta_{\text{init}}(k)$ is the primordial perturbation amplitude (Cos_01 inflation-residual inheritance).

Three growth regimes (standard linear-theory phenomenology, substrate-side reading):

Regime	Era	Substrate-graph content
Radiation suppression	$z > z_{\text{eq}} \approx 3400$	Meszaros effect: sub-horizon matter perturbations frozen by radiation pressure; $\delta \approx \text{const}$
Matter growth	$z_{\Lambda} < z < z_{\text{eq}}$	$D(a) \propto a$; linear growth in matter-dominated regime
Late-LDE suppression	$z < z_{\Lambda} \approx 0.7$	$D(a)$ grows slower than a ; asymptotes as universe approaches de Sitter (Cos_05 saturation regime)

FORM: $P(k, a) = D^2(a)T^2(k)P_{\text{init}}(k)$ — **IDENTIFIED** via Cos_01 + Cos_02 + Cos_03 + Cos_05 + Route A4 + standard cosmological-perturbation-theory composition. Zero pure-D rows.

VALUE: $D(a)$, $T(k)$, $P_{\text{init}}(k)$ all inherit standard cosmology + Route A4 substrate-parameter-INHERITED H_0 ; specific $\sigma_8, P(k)$ values inherit standard CAMB / CLASS Boltzmann-code outputs.

IC: primordial spectrum $n_s \approx 0.965$ + tensor-to-scalar ratio $r < 0.06$ from Cos_01; matter density $\Omega_m h^2 \approx 0.143$, baryon density $\Omega_b h^2 \approx 0.0224$ from Cos_02 + Cos_03 CMB cross-consistency.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime per §4 audit table.

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P03, P09, P10, P11, P12, P13.

Upstream papers: - Paper_012 (P-RB-1 substrate-c) - Paper_027 (Newton's G) - Paper_028 (cosmic decoupling R_H) - Paper_073 (DCGT) - Paper_ED_SC_4_9 (saddle classification) - Paper_ED_Cos_01 (M3 — primordial perturbation spectrum from inflation residual) - Paper_ED_Cos_02 (M3 — baryon density $\Omega_b h^2$ + matter-radiation transition input) - Paper_ED_Cos_03 (M3 — acoustic-physics framework + sound horizon r_s for BAO + transfer-function content) - Paper_ED_Cos_05 (M3 — late-time expansion history + Route A4 H_0 for late-LDE growth-suppression scale) - Paper_ED_Dyn_02 (M2 — RDE/MDE chain-class identification at upstream status; non-load-bearing) - Update_ED_SC_4x_Arc_M3 - Standard cosmological-perturbation-theory machinery (Newtonian/relativistic perturbation theory; gauge formalism; CAMB / CLASS Boltzmann codes for transfer functions + growth factors) — textbook inheritance via DCGT continuum bridge - Standard nonlinear structure formation machinery (halofit; EFTofLSS; N-body simulations) — textbook inheritance (linear-theory only at substrate-graph level here)

Substrate-research support: - Memo_ED_RouteA_A4_Audit + Memo_ED_RouteA_MultiRouteConvergen (Route A4 substrate-parameter-INHERITED closure) - Memo_ED_Q1Q2_JointClosure_Construct + Memo_ED_ChainClass_C3_Construct + Audit

No new substrate primitives. No paper-specific postulates.

§3 Derivation

§3.1 Radiation-era suppression (Meszaros effect)

During the radiation-dominated era ($z > z_{\text{eq}} \approx 3400$), sub-horizon matter density perturbations δ_m satisfy the linearized perturbation equation:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G\rho_m\delta_m = 0$$

In RDE, $\rho_{\text{rad}} \gg \rho_m$ and $H \propto T^2$ (inherits substrate-graph $H(t)$ via Cos_01 + Cos_05 framework + standard cosmology at RDE). The driving term $4\pi G\rho_m$ is subdominant compared to the Hubble friction $2H\dot{\delta}$; matter perturbations grow only logarithmically ($\delta_m \sim \ln a$) rather than as a power law. This is the **Meszaros effect** — matter perturbation suppression during radiation domination.

Super-horizon modes ($k < aH$) grow as $\delta_m \propto a^2$ during RDE (gauge-dependent; in synchronous gauge); these match continuously to sub-horizon evolution at horizon-crossing. Substrate-graph composition: standard linear theory + substrate-graph-inherited $H(t)$ + Cos_02 baryon density at RDE $\rightarrow$ MDE transition.

The matter-radiation equality scale $k_{\text{eq}} \equiv a_{\text{eq}}H_{\text{eq}} \approx 0.01 \text{ h/Mpc}$ sets the turnover in $P(k)$: modes with $k > k_{\text{eq}}$ entered the horizon during RDE (Meszaros-suppressed); modes with $k < k_{\text{eq}}$ entered the horizon during MDE (no Meszaros suppression).

§3.2 Matter-era growth $D(a) \propto a$

Once $a > a_{\text{eq}}$ and the universe is matter-dominated, the growth equation becomes:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}\Omega_m H^2\delta_m = 0$$

In Einstein-de Sitter (EdS; $\Omega_m = 1$) limit, the growing-mode solution is $D(a) \propto a$ (linear growth). For realistic Λ CDM with $\Omega_m \approx 0.31$ today, the growth factor at high z matches EdS ($D \propto a$); at low z (LDE-onset epoch), D slows.

Substrate-graph composition: standard linear theory + substrate-graph-inherited $H(t)$ from Cos_05 expansion history + Cos_02 baryon density + Cos_03 baryon-photon decoupling at recombination.

§3.3 Baryon acoustic imprint on $P(k)$

Pre-recombination, baryon-photon coupling means baryons participate in acoustic oscillations at sound speed c_s (Cos_03 §3.1); these oscillations imprint a characteristic length scale $r_s \approx 147 \text{ Mpc}$ (comoving) on the matter distribution at recombination. After recombination, baryons decouple from photons; the acoustic-feature imprint becomes a **Baryon Acoustic Oscillation (BAO)** signature in the matter power spectrum $P(k)$ as an oscillatory feature centered at wavenumber $k_{\text{BAO}} \approx 2\pi/r_s \approx 0.04 \text{ h/Mpc}$, and equivalently as a peak in the matter correlation function $\xi(r)$ at $r \approx r_s$.

Substrate-graph composition: Cos_03 sound horizon r_s + standard cosmological-perturbation-theory + Cos_02 baryon density $\rightarrow$ BAO feature in $P(k)$. The BAO scale serves as a **standard ruler** for galaxy-survey distance measurements (BOSS, eBOSS, DESI).

Substrate-parameter-INHERITED scaling: $r_s \propto 1/H_0 \propto \Theta_{\text{ED}}^{-1/2}$ via Route A4; observed BAO scale provides H_0 + dark-energy-equation-of-state constraint via galaxy-survey distance-redshift measurements.

§3.4 Transfer function $T(k)$

The transfer function $T(k)$ encodes the relative suppression of small-scale (large- k) modes that entered the horizon during radiation era compared to large-scale (small- k) modes that entered during matter era:

$$T(k) \approx \begin{cases} 1 & k \ll k_{\text{eq}} \\ (k_{\text{eq}}/k)^2 \ln(k/k_{\text{eq}}) & k \gg k_{\text{eq}} \end{cases}$$

(standard analytic approximation; precise $T(k)$ from CAMB / CLASS Boltzmann codes). Combined with primordial spectrum $P_{\text{init}}(k) \propto k^{n_s-1}$:

$$P(k, a=1) = D^2(1) \cdot T^2(k) \cdot P_{\text{init}}(k)$$

gives the present-epoch matter power spectrum with characteristic peak at k_{eq} + small-scale slope $P(k) \propto k^{n_s-4}$ at $k \gg k_{\text{eq}}$ + BAO oscillatory feature.

Substrate-graph composition: Cos_03 acoustic-physics + k_{eq} inheritance from Cos_01 + Cos_02 matter-radiation equality + standard cosmological-perturbation-theory transfer-function calculation.

§3.5 Late-LDE suppression of growth

At late epochs ($z < z_\Lambda \approx 0.7$) when dark energy begins to dominate the cosmological energy budget, the growth factor $D(a)$ grows more slowly than a . Approximate analytic form:

$$D(a) \propto a \cdot g(a), \quad g(a) \rightarrow 1 \text{ at } a \ll a_\Lambda, \quad g(a) \rightarrow \text{const at } a \gg a_\Lambda$$

In the late-LDE / dark-energy-dominated asymptote, growth eventually freezes: $D(a) \rightarrow \text{const}$ as universe approaches Cos_05 saturation regime (de Sitter). Growth-rate parameter $f \equiv d \ln D / d \ln a$ today $\approx \Omega_m^{0.55} \approx 0.5$ — observable via redshift-space-distortion (RSD) measurements in galaxy surveys.

The combination $f\sigma_8(z)$ is the standard structure-formation observable; measurements across $z \in [0, 1]$ from BOSS / eBOSS / DESI provide tight constraints on dark-energy equation-of-state + ED-vs-alternative-cosmology discriminators.

Substrate-graph composition: Cos_05 late-LDE saturation framework + Route A4 H_0 substrate-parameter-INHERITED scaling + standard cosmological-perturbation-theory at LDE. Note on scope: ED supplies no late-LDE growth-suppression mechanism *beyond* standard Λ CDM at this level — the $D(a) \cdot g(a)$ form inherits standard linear theory at LDE; the substrate-graph contribution

is the identification of the LDE-onset epoch with Cos_05's saturation-regime entry plus the Route A4 substrate-parameter scaling of H_0 that sets the onset timing. Consistent with Preamble item 1.

§3.6 Sensitivity to Θ_{ED} or τ_{V_5}

Substrate-parameter Θ_{ED} enters linear-structure observables through Route A4's $H_0 \propto \Theta_{\text{ED}}^{1/2}$:

- $k_{\text{eq}} \propto H_{\text{eq}} \propto H_0$ (matter-radiation equality at fixed z_{eq})
- $r_s \propto 1/H_0$ (BAO standard ruler)
- $D(a)$: shape modified by H_0 -dependent dark-energy onset epoch
- σ_8 : amplitude inherits Cos_01 primordial-spectrum normalization $\times D(a)^2 \times T(k)^2$ chain

$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ **tension framing (honest scope)**: CMB-derived $S_8^{\text{CMB}} \approx 0.834 \pm 0.016$ (Planck) vs weak-lensing-derived $S_8^{\text{WL}} \approx 0.766 \pm 0.020$ (KiDS-1000 + DES Y3) — ~ 3 discrepancy as of 2024. Structurally parallel to the H_0 Hubble tension (Cos_03 §3.6): standard ΛCDM also predicts a single consistent S_8 under a given parameter set — the tension is between measurement channels of the same parameter, not between two parameter values, and ED's Route A4 channel does not naturally resolve it. **Route A4 supplies ED's structural link to the S_8 -tension observational target via the substrate-parameter scaling channel**; whether the tension resolves via substrate-parameter tightening (RA-OPEN-4a closure) or standard intermediate-redshift physics modification (early dark energy, baryonic feedback at small scales — standard-cosmology / astrophysics territory) is OPEN and **not load-bearing for Cos_04's M3 verdict**. Either pathway requires substrate-research-frontier or standard-cosmology-level work; neither aggravates the Cos_04 chain.

Tightening via RA-OPEN-4a/4b/4c-explicit substrate-research-frontier, not blocking for Cos_04's M3 verdict.

§3.7 Comparison to ΛCDM linear structure theory

Element	ΛCDM standard	ED substrate-graph reading	Status
Growth factor $D(a)$ from linear perturbation equation	Standard Friedmann + linear theory	Substrate-graph $H(t)$ via Cos_01 + Cos_05 + Route A4 inheritance	D-via-I via continuum-Friedmann inheritance
Meszaros radiation-era suppression	Standard linear theory	Same; substrate-graph $H(t)$ at RDE	D-via-I
Matter-era growth $D(a) \propto a$	Standard EdS-limit growing mode	Same; substrate-graph $H(t)$ at MDE inherits Cos_05 + standard cosmology	D-via-I
Late-LDE growth suppression	Standard dark-energy phenomenology	Cos_05 saturation framework + Route A4 H_0	D-via-I
BAO standard ruler at $r_s \approx 147$ Mpc	Standard acoustic-imprint	Cos_03 sound horizon inheritance	D-via-I

Element	Λ CDM standard	ED substrate-graph reading	Status
Transfer function $T(k)$ with k_{eq} turnover	Standard cosmological-perturbation theory	Cos_03 acoustic-physics + Cos_01 + Cos_02 inheritance	D-via-I
Power spectrum $P(k, a) = D^2 T^2 P_{\text{init}}$	Standard linear-theory composition	Same; substrate-graph composition of upstream Cos_x papers	D-via-I
$f\sigma_8(z)$ growth-rate observable	Standard RSD measurement target	Same; substrate-graph framework via Cos_05 + standard linear theory	D-via-I
Substrate-parameter Θ_{ED} sensitivity	None	$H_0 \propto \Theta_{\text{ED}}^{1/2}$ via Route A4 $\rightarrow$ linear-structure observables inherit	Distinguishing content
S_8 tension (CMB vs weak-lensing)	Observational puzzle	Same; substrate-research-frontier tightening direction	Inherited puzzle + distinguishing content via Route A4

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	P	Paper_087.
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_028; Paper_073; Paper_ED_SC_4_9; Paper_ED_Cos_01 (M3); Paper_ED_Cos_02 (M3); Paper_ED_Cos_03 (M3); Paper_ED_Cos_05 (M3); Paper_ED_Dyn_02 (M2 — upstream, non-load-bearing) ; Up- date_ED_SC_4x_Arc_M3.

#	Step	Label	Notes
3	Standard cosmological-perturbation-theory machinery (Newtonian / relativistic; gauge formalism)	I	Textbook inheritance via DCGT continuum bridge.
4	Standard Boltzmann codes (CAMB / CLASS) for transfer functions + growth factors	I	Textbook inheritance for specific $D(a)$, $T(k)$, $P(k)$ outputs.
5	Standard nonlinear / N-body machinery (halofit / EFTofLSS)	I	Textbook; substrate-graph framework supplies linear-theory only here.
6	Meszaros radiation-era suppression: $\delta_m \sim \ln a$ for sub-horizon modes	D-via-I	§3.1. Composition of standard linear theory + substrate-graph $H(t)$ at RDE.
7	Matter-era growth $D(a) \propto a$ in EdS limit	D-via-I	§3.2. Standard linear theory + substrate-graph $H(t)$ at MDE via Cos_05.
8	Matter-radiation equality scale $k_{\text{eq}} \approx 0.01 \text{ h/Mpc}$ as turnover in $P(k)$	D-via-I	§3.1, §3.4. Cos_01 + Cos_02 + standard cosmology.
9	BAO standard ruler at $r_s \approx 147 \text{ Mpc}$ imprinted on $P(k)$	D-via-I	§3.3. Cos_03 sound horizon + standard linear theory + Cos_02 baryon density.
10	Transfer function $T(k)$ with $T \rightarrow 1$ at $k \ll k_{\text{eq}}$, suppressed at $k \gg k_{\text{eq}}$	D-via-I	§3.4. Cos_03 acoustic physics + standard cosmological-perturbation theory.
11	Primordial spectrum $P_{\text{init}}(k) \propto k^{n_s-1}$ with $n_s \approx 0.965$	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Cos_01 inflation-residual + standard inflationary-perturbation theory.

#	Step	Label	Notes
12	Full matter power spectrum $P(k, a) = D^2(a)T^2(k)P_{\text{init}}(k)$	D-via-I	§3.4, §3.7. Composition of rows 7, 10, 11.
13	Late-LDE growth suppression $D(a) \cdot g(a)$ from Cos_05 saturation regime	D-via-I	§3.5. Cos_05 late-LDE framework + standard linear theory.
14	$f\sigma_8(z)$ growth-rate observable for RSD measurements	D-via-I	§3.5. Standard linear theory + Cos_05 inheritance.
15	Substrate-parameter Θ_{ED} sensitivity: $k_{\text{eq}}, r_s, H_0 \propto \Theta_{\text{ED}}^{1/2}$ scaling	D-via-I	§3.6. Composition of Route A4 + standard linear-structure observables.
16	Substrate-c bound preserved throughout	D-via-I	§3.1. Paper_012 + Paper_089.
17	H_0 at Route A4 substrate-parameter-INHERITED level	I	Sub-type per Paper_095 §3.2.1: <i>substrate-parameter-INHERITED</i> . §3.5, §3.6. Paper_027-analog tier per Memo_RouteA_MultiRouteConvergence option (ii).
18	Matter density $\Omega_m h^2$, baryon density $\Omega_b h^2$ from CMB + BBN cross-consistency	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Cos_02 + Cos_03 cross-consistency.
19	Recombination + post-recombination matter dynamics from Cos_03 + standard cosmology	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Cos_03 acoustic-physics framework.
20	S_8 tension (CMB vs weak-lensing): substrate-research-frontier tightening direction	OPEN (non-load-bearing)	§3.6. Inherited observational puzzle.

#	Step	Label	Notes
21	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime	A→position	Per Paper_095 §3.3 line 77. Five-anchor verdict-sync verified.

Audit summary: zero pure-D rows; **ten substantive D-via-I rows** (6-10, 12-16); five I rows (2, 3, 4, 5, 17); three IC-INHERITED rows (11, 18, 19); one non-load-bearing OPEN (20 — S_8 tension); one P row (1); one A→position verdict row (21). **No load-bearing OPEN items.** Upstream Dyn_02 M2 inherited at upstream status without load-bearing. No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest):** Observed matter power-spectrum $P(k)$ shape or BAO scale significantly inconsistent with substrate-graph composition $D^2(a)T^2(k)P_{\text{init}}(k)$ under Route A4 H_0 substrate-parameter scaling — beyond galaxy-survey + CMB-lensing observational precision ($\sim 1\%$ on BAO scale; few-% on $P(k)$ shape). Refutes §3.3-§3.4 + audit rows 9, 12. Currently consistent: BOSS / eBOSS / DESI BAO + DES / KiDS / HSC weak-lensing + Planck CMB-lensing match Λ CDM at percent precision.
- **F2:** Observed $f\sigma_8(z)$ growth-rate measurements from RSD significantly inconsistent with substrate-graph $D(a)$ inherited from Cos_05 + standard linear theory. Refutes §3.5 + audit row 14.
- **F3:** Observed $P(k)$ slope significantly inconsistent with substrate-graph transfer function $T(k)$ + primordial spectrum n_s inheritance. Refutes §3.4 + audit rows 10, 11, 12.
- **F4 (substrate-research-frontier-conditional):** Substrate-graph proof that Route A4 H_0 substrate-parameter scaling produces structurally inconsistent $r_s, \sigma_8, \Omega_m h^2$ predictions when simultaneously constrained by Planck TT/TE/EE + BAO + weak-lensing at observed precision. **Empirical refutation not currently available** (depends on RA-OPEN-4a closure + S_8 -tension observational resolution). Refutes §3.6 + audit rows 15, 17.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

Linear structure formation in ED is the substrate-graph reading of standard linearized cosmological-perturbation theory on the ED expansion-history background: growth factor $D(a)$ from Cos_05 + Cos_01 + Route A4 inheritance; transfer function $T(k)$ from Cos_03 acoustic-physics

+ Cos_02 baryon density; primordial spectrum from Cos_01 inflation residual; BAO standard ruler at $r_s \approx 147$ Mpc from Cos_03 sound horizon. Full matter power spectrum $P(k, a) = D^2(a)T^2(k)P_{\text{init}}(k)$ inherits standard cosmological-perturbation-theory composition. Substrate-c bound preserved throughout.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Ten D-via-I rows; five I rows; one substrate-parameter-INHERITED row (Route A4); three IC-INHERITED rows (primordial spectrum + matter/baryon densities + recombination); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items**. One non-load-bearing OPEN (S_8 tension between CMB and weak-lensing; inherited observational puzzle with substrate-research-frontier tightening direction). Upstream Dyn_02 M2 (RDE/MDE chain-class identification) inherited at upstream status; not load-bearing for Cos_04 net M3 via continuum-Friedmann inheritance.

Empirical-anchor strength: BOSS / eBOSS / DESI BAO at sub-percent precision; DES / KiDS / HSC weak-lensing; Planck CMB-lensing; future Euclid / LSST / Roman precision — collectively one of the most data-rich observational sectors in cosmology. The **distinguishing ED contribution** is the Route A4 substrate-parameter Θ_{ED} scaling channel for *all* linear-structure observables ($D(a)$, $T(k)$, r_s , σ_8 , $f\sigma_8$) — a single substrate-parameter knob threading through every linear-theory output. The S_8 tension observational target (and the parallel Hubble tension via Cos_03) inherits this scaling channel as ED’s structural link to the tension, without claiming substrate-side resolution.

Cross-arc consequences. Upstream, Cos_04 inherits unchanged from Cos_01 (M3), Cos_02 (M3), Cos_03 (M3), Cos_05 (M3), Dyn_02 (M2 — chain-class OPENs not aggravated per Preamble item 6). Downstream, Cos_04 *enables* Paper_ED_Cos_06 (Inflationary fluctuation spectrum) via composition with Cos_01 + standard inflationary-perturbation theory (verdict tier depending on Cos_06’s own chain — not forecast here); *strengthens* the SCBU cosmology projection via BAO + weak-lensing + CMB-lensing empirical anchors at substrate-parameter-INHERITED level; *strengthens* the empirical-prediction sector through the Route A4 S_8 -tension structural-link channel; *strengthens* future-observatory observational anchors (DESI ongoing; Euclid post-2024 launch; LSST / Vera Rubin; Roman Space Telescope) as ED-vs-alternative-cosmology discriminator channels via the Route A4 scaling. **Cosmology Arc completion milestone:** Cos_01 (Inflation, M3) + Cos_02 (BBN, M3) + Cos_03 (CMB Acoustic, M3) + Cos_04 (this paper, M3) + Cos_05 (Dark Energy, M3) + Cos_06 (Inflationary Spectrum, M3) + Paper_038_5 (Λ -smallness reframing, **M3 retroactive** — upgrade applied 2026-05-17 via Route A4 substrate-parameter-INHERITED closure) + **Paper_ED_Baryogenesis (M3 — upgraded 2026-05-17 from M2 via R4 single-template admissibility + IC-supply equivalence per Update_Paper_ED_Baryogenesis_M3_via_R4)** form the substantively-complete Cosmology Arc spanning primordial perturbations → nucleosynthesis → recombination → linear structure → dark-energy-dominated late universe + Λ -smallness reframing + matter-antimatter asymmetry origin. **Eight Cosmology Arc papers at M3.**

End Paper_ED_Cos_04.